PRASANTH RAYUDU

GPU Engineer — GPU ASIC Architecture, Functional Modeling, Performance & Power Verification

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PROFESSIONAL SUMMARY

Computer Engineering graduate student with an Electrical Engineering and foundation, committed to architecting, designing, implementing, verifying, and optimizing GPU ASIC cores for performance and power, backed by three-plus years across FPGA, RTL, and GPU-adjacent architecture modeling roles. Builds functional modeling simulations and Python/C++ software, tools, and tests for graphics and compute workloads, runs benchmarks across applications, and performs pre- and post-silicon verification to confirm correctness and meet performance and power goals. Proficient in SystemVerilog/Verilog, C/C++, Python, MATLAB, Tcl, and Bash, with regular participation in design reviews and cross-functional project meetings alongside architecture, design, verification, firmware, and software teams. Experienced with Xilinx UltraScale/MPSoC platforms, Synopsys VCS, ModelSim, Vivado, waveform debugging, timing closure, clock-domain crossing, PPA optimization, and parallel-compute workload analysis.

CORE SKILLS

GPU ASIC / Architecture Modeling: Functional modeling simulations, graphics and compute workload benchmarking, performance analysis, power modeling, design-space exploration, PPA tradeoffs, parallel-compute concepts, GPU core architecture, GPU ASIC design

Verification (Pre- and Post-Silicon): SystemVerilog, Verilog, VHDL, UVM, RTL design, testbench development, BFM/transactors, coverage monitors, static timing analysis (STA), pre-silicon validation, post-silicon bring-up correlation, FPGA emulation, synthesis, timing closure, CDC

Programming and Automation: C, C++, Python, MATLAB, Tcl, Bash/Shell, Make, CMake, Git, CI/CD, log parsing, benchmark automation, regression automation, test development

Tools and Platforms: Synopsys VCS, ModelSim, Xilinx Vivado, Xilinx ILA, UltraScale/UltraScale+, MPSoC, Quartus/Intel FPGA basics, Linux, Embedded Linux

Interfaces and Systems: AXI, PCIe, I2C, LVDS, Ethernet, memory-mapped I/O, interrupts, ARM-based systems, hardware-software co-simulation

Power and Performance: VCD activity analysis, switching-factor modeling, clock gating, DVFS awareness, energy/latency optimization, workload metrics, performance-per-watt analysis

WORK EXPERIENCE

Graduate Research Assistant — Hardware-Software Co-Design for Edge AI, San Diego State University **2025 - Present**

- Developed HyperOPT, a hardware-software co-design framework and functional modeling simulation that explores algorithm and accelerator configurations under accuracy, area, latency, and energy constraints — directly analogous to GPU core performance/power design-space exploration.
- Built Python-based functional models, software tools, and automated test pipelines for HDC compute workloads mapped to compute-in-memory accelerators; ran benchmarks across 320 design combinations and identified 19 feasible candidates, achieving 91.14% accuracy at 1,354 pJ compute energy and meeting defined performance and power goals.
- Modeled quantization, analog noise, memory-cell non-idealities, ADC overhead, and multi-macro mapping to quantify performance-per-watt and accuracy tradeoffs; verified proxy estimates against detailed architecture-level results and presented findings at design reviews.
- Implemented pruning and binary-search strategies to accelerate design-space exploration and produce reproducible benchmark reports for design reviews and project meetings.

FPGA Emulation Flow Automation with CI/CD Integration **2024 - Present**

- Built a scalable FPGA emulation flow supporting remote access, CI-triggered builds, regression execution, and simulation-to-hardware handoff on Xilinx UltraScale-class platforms for pre- and post-silicon verification.
- Developed Python, Tcl, Make, and Bash software and test automation for compilation, environment configuration, workload launch, benchmark execution, metric collection, and artifact publishing; added log parsing and failure clustering enabling root-cause triage for more than 95% of simulation-to-emulation mismatches.
- Integrated Git-based configuration management and concurrent test branches, reducing compile/debug iteration time by 30%; collected performance and power counters across regression workloads to verify performance and power goals were met.

Teaching Associate — General Studies, San Diego State University **2024 - 2025**

- Led weekly first-year seminar sections, facilitated technical and academic discussions, tracked student progress, and delivered timely assessment through Canvas LMS.
- Communicated complex material to diverse audiences and collaborated in weekly teaching communities to standardize course execution and improve student outcomes.

Student Intern — FPGA and Machine Learning Systems, In22 Labs **2021 - 2022**

- Designed and validated Verilog RTL blocks for a biomedical machine-learning pipeline and co-simulated embedded C reference models to verify functional correctness.

- Automated ModelSim waveform and log validation in Python, reducing manual debug effort by 50% and improving regression reproducibility.
- Collaborated with FPGA, software, and data-science engineers to reduce end-to-end inference latency and support ARM-based hardware-software integration.

SELECTED ENGINEERING PROJECTS

System-Level FPGA Emulation Framework for SoC Subsystem Validation **2023 - 2024**

- Integrated a multi-IP CPU-microcontroller subsystem on a Xilinx MPSoC FPGA for pre-silicon validation and ARM embedded-Linux bring-up, supporting post-silicon correlation.
- Created reusable SystemVerilog testbenches, BFMs, and transactors for AXI, interrupts, PCIe, I2C, and LVDS; automated compile/runtime flows with Tcl and Make, cutting build cycles 40% (Synopsys VCS, ModelSim, Vivado, Xilinx ILA).

GPU Workload Modeling & Power-Performance Analysis **2023 - 2024**

- Built functional simulations and modeling infrastructure to assess GPU subsystem workloads and execution behavior using synthetic and real-world compute kernels.
- Developed automated Python and C++ scripts to run power/performance benchmarks on simulated RTL and post-silicon platforms, achieving a 15% improvement in power estimation accuracy.
- Implemented regression test suites and coverage monitors using SystemVerilog and UVM to validate GPU feature blocks across multiple scenarios.
- Performed PPA (Power, Performance, Area) trade-off analysis on candidate microarchitectures, optimizing power and performance and contributing to design choices adopted in final GPU tapeout.

RTL Design and Verification of Shader Processing Unit (SPU) **2022 - 2023**

- Designed and verified RTL modules for a GPU Shader Processing Unit, including control logic and data path for texture and math engines, using Verilog and SystemVerilog.
- Developed timing-accurate simulations and resolved clock-domain crossing (CDC) issues during integration into the SoC simulation environment.
- Conducted static timing analysis (STA) and achieved timing closure on a Xilinx MPSoC FPGA prototype using Vivado.
- Collaborated with architecture and ASIC teams to ensure compliance with performance and power targets under gaming and AI workloads.

EDUCATION

Master of Science, Computer Engineering — San Diego State University (SDSU) **Expected Fall 2026**

Relevant Coursework: Computer Architecture, VLSI Design, Digital Systems, Embedded Systems, Wireless Communication Systems, RF Circuit Design, Electromagnetics, Antenna Theory, High-Speed PCB Design

Bachelor of Technology, Electronics and Communication Engineering — VR Siddhartha Engineering College